MATTHEW PORCELLI

Scottsdale, AZ | 480-212-2274 | MattPorcelli@Protonmail.com
https://mattporcelli.github.io/portfolio/

PROFESSIONAL SUMMARY

Senior Network Engineer with 20+ years in telecommunications, including 17+ in service-provider network engineering at Cox Communications, specializing in IP/MPLS, DWDM/optical transport, QoS design, and network automation. Sets national engineering standards for the Cox Business Network, leads High-Level Design (HLD) for Nokia optical and IP/MPLS deployments, and owns capacity planning and lab/lifecycle budgeting. Builds network automation and lab tooling (Containerlab, Docker, Python, gNMIc, Prometheus/Grafana, MCP servers) adopted across engineering teams. Hands-on with Nokia 7x50/SR, Juniper MX, Cisco ASR/CRS, and Cisco Meraki platforms; BGP, IS-IS, OSPF, Segment Routing (SR-MPLS), and Carrier Ethernet (MEF).

CORE COMPETENCIES

Network Design & Planning: IP/MPLS Designs • DWDM/Optical Designs • QoS Strategy • Capacity Planning • Requirements-Based Hardware Selection • Production & Lab Budgetary Planning • Lifecycle Management

Automation & Platforms: Network Automation • Cisco Meraki • CI/CD Pipelines • REST APIs • MCP Servers • Tooling Development • Containerlab • Docker • Python • Linux

Testing & Observability: Prometheus • Grafana • Observability • Test Automation • Robot Framework • Regression Testing • Validation & Rollout Testing • Pre/Post-Check Automation

Governance & Documentation: National Market Governance • Engineering Standards • Technical Documentation • Methods of Procedure (MOPs) • Stakeholder Communication • Change Engineering & Validation

PROFESSIONAL EXPERIENCE

Cox Communications (Cox Business) — 2009 – Present

Sr Communications / Network Engineer III — National Engineering Design — 2020 – Present

- **Optical Network Design Leadership:** Lead Design Engineer for Nokia Optical projects (1830 PSS platform: PSS-16II, PSS-8, PSD-10, PSD-2) within the Cox Business Network, nationally, covering DWDM design and deployment across 3 regions and 11 markets. Lead design for the Access network (600+ nodes, 1,200+ services) and design across the Core (1,100+ nodes, 2,700+ services) and OTN (200+ nodes, 2,000+ services). Evaluate new optical hardware and bring it into lab evaluation to validate capabilities and fit before adoption.
- **IP/MPLS Design Standards:** Set the national IP/MPLS design and configuration standards used to build and deploy services across the Core (700+ routers) and Access (79,000+ routers) layers of the Cox Business Network (1.2M+ services network-wide), including design and lab testing of Nokia IXR-e2 and IXR-e2c platforms.
- **Meraki SD-WAN Underlay Design:** Contributed to design of the underlay network product for the Cox Business SD-WAN service offering built on Cisco Meraki, collaborating to understand the overlay's needs and the requirements and class of service (CoS) the product required. Configured the lab environment and performed lab testing, including Meraki high-availability failover scenarios, to validate the design.
- **NOCaaS Alarm Monitoring Proof of Concept:** Conducted proof-of-concept testing using Cisco Meraki gear to evaluate it as a potential replacement for NOCaaS (NOC as a Service) customer network alarm monitoring; testing showed the hardware lacked specific required capabilities.
- **Non-BAU Design & Documentation:** Engineering resource for non-BAU optical and IP/MPLS deployments requiring unique capacity or design considerations. Author High-Level Design (HLD) documents for IP/MPLS and

DWDM technology evaluations and lab testing initiatives to guide adoption decisions across the Cox Business Network.

- **Technology Evaluation & Validation:** Evaluate new platforms and features, including EVPN over SR-MPLS with SR-TE in Containerlab lab topologies. Combine technical assessment with lab and regression testing, where applicable, to validate capabilities, limitations, network integration, release stability, and cost considerations, supporting requirements-based hardware selection. Document findings in an evaluation document that is handed off to inform adoption decisions.
- **Budgetary, Capacity & Lifecycle Planning:** Lead optical gear lifecycle budgeting, including optical burden calculations across network layers. Own production and lab budgeting for optical and IP/MPLS equipment, driving cost savings by virtualizing lab infrastructure with Containerlab.
- **Automation, Lab Infrastructure & Observability:** Architect and maintain Containerlab environments and VM/Docker infrastructure (10 VMs, 25+ containers) on RHEL 8/9 and Rocky Linux for network testing and R&D, including containerized lab services (TACACS+, Kea DHCP). Deploy gNMIc streaming telemetry with Prometheus and Grafana for lab observability and visualization. Run Docker image builds through Jenkins CI/CD pipelines. Maintain lab drawings and design documentation for all new and existing lab devices (30+ core and 100+ access IP/MPLS routers, 20+ transport access nodes), providing the accurate device inventory that drives lab budgetary and capacity planning.
- **Custom Application Development:** Leveraged AI-assisted development tools to design and build:
  - **Harbor:** Unified hardware lab testing workspace that integrates engineering tools through REST APIs so engineers can work without switching between separate applications. Docker containers connect Robot Framework tests to Ixia traffic generators, run automated pre/post-change checks, analyze test traffic, and report findings.
  - **CLAB Orchestrator:** Drag-and-drop Containerlab topology builder integrating gNMIc, Grafana, Prometheus, and Wireshark.
  - **ZTP Orchestrator (proof of concept):** Zero-touch provisioning driven by DHCP Option 67 on a containerized Kea DHCP server.
  - **Endpoint:** Nokia WSNOC API-driven hardware lifecycle and budget tracking.
- **MCP Server Development:** Designed and deployed independent MCP servers, each containerized per application (Nokia WSNOC (formerly NFM-T), NFM-P/NSP, Elastic, Snowflake, CB-NEAT) and integrated through REST APIs, with tools scoped specifically to each system.

Sr Communications / Network Engineer III — National Sustaining Engineering 2018 – 2020

- **QoS Design & Strategy:** Owned QoS design, strategy, and policy for the Cox Business Network, setting policer and shaper QoS policy standards adopted across the network. Documented standards in the "QoS Standards Guide" and built QoS calculation spreadsheets for reporting.
- **Tier IV Escalation Authority:** Served as the highest tier of escalation support (Tier IV), engaged when an issue went beyond routine troubleshooting and standards-level expertise was needed to resolve it. Successfully resolved major failed-SLA/circuit issues for large customers, including Sprint (45+ circuits) and T-Mobile (85+ circuits), along with IPVPN MTU failures, multicast QoS policies failing above 2 Gbps, and 7250 IXR frame discards. Produced Technical Bulletins and a QoS Decision Tree adopted across 3 regions and 11 markets.
- **Video Product Engineering:** Drove Pro Idiom (multicast video for Cox Business/Hospitality) and Contour for Hospitality (unicast IPTV replacement) by engineering the underlay network service delivering both products. Developed and lab-tested CPE-less design standards (eliminating the customer-premises device), co-authored national playbooks with Atlanta Video/Metro Engineering and OSS, and led training across 3 regions and 11 markets.

- **RSAAT Test Automation Platform (Visionworks/Spirent):** Owned the Remote Service and Activation Testing (RSAAT) platform end-to-end, defining requirements through playbook scenarios and testing each capability once built:
  - **Spirent Connectors:** Playbook scenarios defining how the Centralized and MC-LAG connectors work for automated circuit test rebuilding.
  - **Spirent Orchestrator:** Design and playbook scenarios for a new 100G-capable Orchestrator (100 simultaneous tests, workflow automation, standardized across markets).
  - **Probe Standardization:** Helped retire end-of-support probes and standardized 100+ probe deployments across all markets and regions.
  - **Remote Packet Capture:** Developed MOPs used by markets for customer integration troubleshooting.
- **Engineering Standards & Documentation:** Authored and governed engineering standards documentation. Authored infrastructure (7750 SR-X Deployment Guide), services (QoS Standards Guide, Pro Idiom & Contour playbooks), process (QoS distribution via NSP, Visionworks/Spirent user provisioning), and testing (Y.1564 SOP, remote packet capture, Spirent connectors) documentation.
- **Cross-Market Collaboration:** Led biweekly Sustaining Collaboration calls and monthly Service Activation & Assurance Testing calls across 3 regions and 11 markets, working with market Field Service Activation (FSA) and Service Assurance (SA) subject-matter experts.

Network Operations Engineer II — Southwest Region **2009 – 2018**

- **Architecture & Design Engineering (Arizona & Las Vegas):**
  - **Capacity Planning:** Planned capacity and presented budget requirements for network growth across the Nokia Core MPLS network (75+ core routers) and access-facing MPLS rings (4,000+ devices).
  - **Baseline Configurations:** Wrote and maintained standard baseline router configurations so deployments were consistent across all sites.
  - **Platforms:** Nokia/Alcatel-Lucent 7x50/7210 SAS-M/7210 MXP/7210 SAS-K/7705 SAR; Juniper MX80; Cisco 3750; Accedian MetroNID/MetroNODE.
- **Compliance & Deployment Standards:** Engineered network changes to meet Granite (circuit inventory accuracy) and Telcordia (telecommunications industry standards) compliance, identifying out-of-compliance network devices and producing engineering remediation packages for Operations teams. Created deployment guides (7210 SAS-K, Y.1731, DS3 circuit emulation with Digital Access and Cross-Connect System (DACS) integration) used across the Southwest Region and validated changes in a lab before production rollout. The DS3 circuit emulation guide was adopted by the national team as a viable way off SONET.
- **RFP Design Leadership:** Lead designer for the Arizona State University and Raiders Stadium (Las Vegas) RFPs, designing point-to-point transport and the IP/MPLS network, and defining how they would be built and leveraged to meet each customer's needs, with the services riding the network designed to engineering standards.
- **Cisco DWDM Networks (Arizona & Las Vegas):** Planned capacity and budget requirements for 40/80/96-channel lambda networks (30+ nodes across 5 networks in the Southwest Region). Designed DWDM circuits through the Metro Core for Cox Business/Cox Communications Video Services, and designed a 96-channel Cisco Flex Spectrum network in Phoenix Metro (2017), training engineers (mostly Tech Ops) and the designers who would work on it.
- **Deployment & Support (Arizona & Las Vegas):**
  - **IP Backbone & Routing:** Configured and supported IP backbone equipment with multi-protocol routing (BGP, IS-IS, OSPF, MPLS) and MEF circuit types (EPL, EVPL, EP-LAN, EVP-LAN) across Cisco 7609/7604/CRS/ASR9000, Cisco 3400/3700, Juniper MX80, and Nokia/Alcatel-Lucent 7x50/7250/7210 platforms.
  - **Voice & TDM Transport:** Configured and troubleshot H.323/SIP voice services (Cisco AS5400/IAD2400, Edgewater ESBCs, Acme SBCs, BroadWorks Softswitch, Nortel DMS-500) and SONET/DWDM transport (Fujitsu Flashwave/Cisco ONS ADMs, Tellabs Titan/NGX cross-connects, SONET DS1–OC192, DWDM 1G–100G).

CERTIFICATIONS

- Nokia Network Routing Specialist I (NRS I) — 2016
- Cisco CCNP Route & Switch, SWITCH exam (300-115) — 2015
- Cisco CCNA — 2005 (recertified 2012)

EDUCATION

- Valencia Community College — A.S., Computer Engineering Technology (Cisco Specialization), 2004
- Valencia Community College — A.A., 2002
- Cisco Networking Academy — Graduated 2004